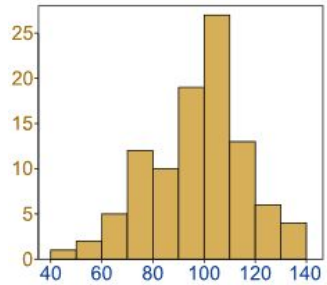
Histogram: A graphical display of data using bars of different heights.

- Each bar of a histogram graphs numbers into ranges
- Histograms display the shape and spread of continuous sample data



This is what a histogram looks like:

The horizontal axis of histograms represent range/intervals

Here's an example of a histogram problem:

- Given the values of:

36	25	38	46	55	68	72	55	36	38
67	45	22	48	91	46	52	61	58	55

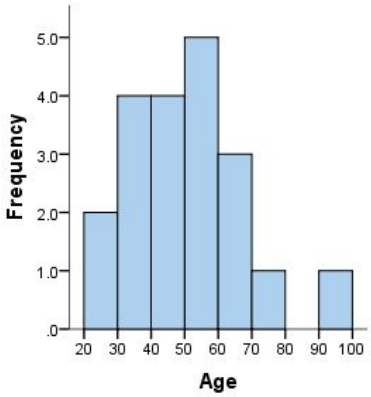
- And an interval of 10 starting from the number 20 to the number 11

We can put the values into a frequency table to see how many values fall into each interval. (For example, there are 2 values that fall into the interval of 20-30. The 2 numbers are 25 and 22)

Interval	Frequency	Scores Included in Interval
20-30	2	25,22
30-40	4	36,38,36,38
40-50	4	46,45,48,46
50-60	5	55,55,52,58,55
60-70	3	68,67,61
70-80	1	72
80-90	0	-
90-100	1	91

The number of values that falls into each interval is the frequency.

The frequency is on the y-axis of the histogram while the interval is on the x-axis of the histogram



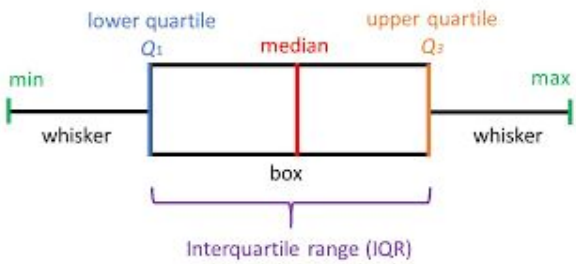
This is what the histogram of this problem looks like

The histogram visualizes the data in the frequency table. (For example, the interval of 20-30 has a frequency of 2, so the bar of the histogram representing the interval of 20-30 is graphed up to the number 2 on the y-axis.)

Problem Credit: https://statistics.laerd.com/statistical-guides/understanding-histograms.php

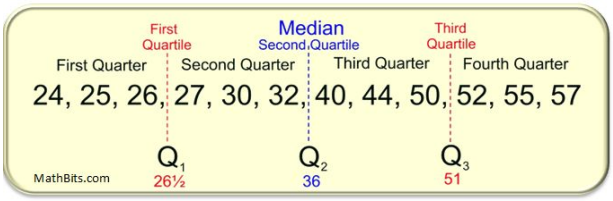
Box Plot: A box plot is a diagram that displays the distribution of data based on the five-number summary. The five-number summary is the minimum, first quartile, median, third quartile, and maximum.

This is what a box plot looks like:



Given a data set:

- The **minimum** is the smallest number
- The **maximum** is the largest number
- The **median** (aka Q2)is the middle number. If there are 2 numbers in the middle, the median is the average of the 2 middle values. 50% of the data lies below the median and 50% lies above
- The **Q1** (aka lower quartile or first quartile) is the middle number between the median and the minimum value. 25% of the data lies below Q1 and 75% lies above
- The **Q3** (aka upper quartile or third quartile) is the middle number between the median and the maximum value. 75% of the data lies below Q3 and 25% lies above
- The **interquartile range** is the difference between Q3 and Q1. [IQR=Q3-Q1]



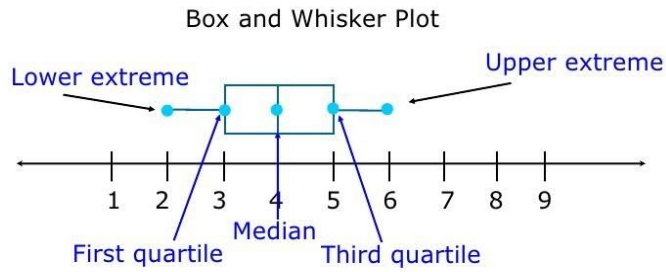
To make a box and whisker plot:

- We must first put the data set in order from the smallest to the largest
- Then, we need to figure out the minimum, first quartile, median, third quartile, and maximum of the data set.
- After that, we mark off the 5 values on a number line
- We put a box around Q1-Q3 with a line inside the box showing Q2
- We then connect the box with the minimum and maximum values, creating 2 whiskers

Example problem:
Values/Data given: 4, 2, 3.5, 5.2, 6, 4.7, 3.6, 5, 3

Steps:

1. Put the values in order from least to greatest: **2, 3.5, 3, 3.6, 4, 4.7, 5, 5.2, 6**
2. Find the 5 values (min: 2, Q1: 3, median: 4, Q3: 5, max: 6)
3. Plot the 5 values on top of a number line
4. Connect the different values as shown using a box and whisker:



This is what the box plot should look like:

The **mean** of a data set is the average of all numbers

- <u>To find the mean</u>
    1. Add up all the numbers
    2. Divide the sum by how many numbers there are

Example:
<u>Given the values:</u> 1, 4, 5, 9, 2
We can find the mean by first adding the numbers together (1+4+5+9+2)=21
Then we divide the sum (21) by the number of values (5). 21/5= 4.2
So, the mean is 4.2

Standard Deviation is a measure of how far the numbers in a data set vary from the mean

- It shows how spread out numbers are
- The symbol for standard deviation is σ

Steps for Calculating Standard Deviation:

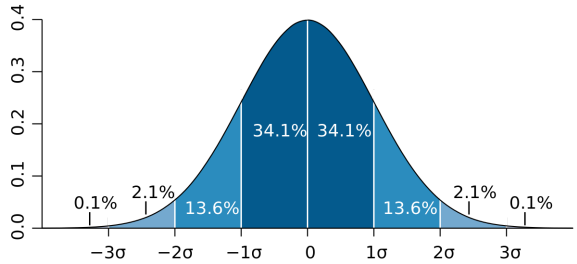
1. Find the mean
2. Subtract the mean from each number in a given data set and square the result
3. Find the average of all the squared differences you found out in step 2
4. Find the square root of the average from step 3

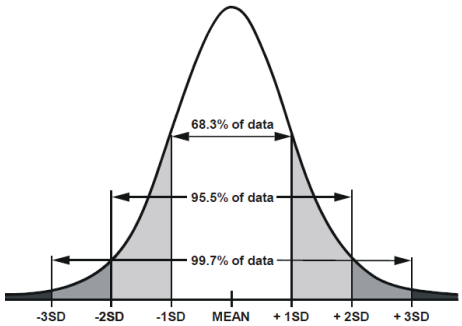
Outlier: A value that is much smaller or larger than other values in a set of data.

- An outlier can skew the results of data analysis. It can significantly affect the mean and the standard deviation of data.

Example:
In a data set of: 12, 15, 18, 20, 13, 17, 51, 14, 11, 19
The number 51 is an outlier

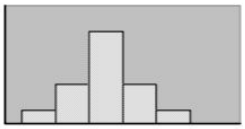
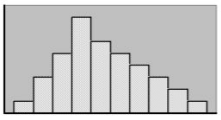
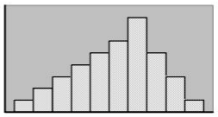
Normal distribution is a data distribution where the data tends to be around a central value with no bias left and right. It is shaped like a bell curve





In a normal distribution:

- About 68% of data fall within one standard deviation of the mean
- 95% of data fall within two standard deviation of the mean
- 99.7% of data fall within three standard deviations of the mean

This is an example of a normal distribution:	**These are 2 examples of skewed distributions:**
 (scores on a "normal" test)	 Right side of the curve is stretched or skewed right  Left side of the curve is stretched or skewed left Outliers can lead to skewed distributions of data